Macroeconomics in the Sequence Space

Chen Gao Weigao Wang

School of Economics, Peking University

April 13, 2026

Outline

1. Sequence Space vs. State Space
2. SSJ Methods
3. SSJ development
   - 3.1 Application
4. Global Solution
   - 4.1 Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function
   - 4.2 DL in sequence space
   - 4.3 RL in sequence space
5. Reference

Sequence Space vs. State Space

Sequence Space vs. State Space

Motivation

- In heterogeneous-agent (HA) models, the economy's state is the entire cross-sectional distribution — an infinite-dimensional object.
- Sequence-space methods *sidestep* this challenge: they characterize local dynamics around the steady state without ever carrying the distribution as a state variable.
- The key insight: aggregate equilibrium responses to shocks can be encoded in Jacobian matrices of aggregate outcomes with respect to aggregate inputs.

Sequence Space vs. State Space

State Space

Definition 1

A state-space representation describes the economy recursively. At date t, a state variable x_t summarizes all payoff-relevant history, and the model evolves as

$$x_{t+1} = f(x_t, \varepsilon_{t+1}), \qquad y_t = g(x_t).$$

Given x_t, the future is independent of the past.

Definition 2

A sequence-space representation characterizes equilibrium as a mapping between entire time paths. We seek sequences $\{X_t\}_{t=0}^{\infty}$ such that equilibrium conditions hold at every date:

$$F(\{X_t\}_{t=0}^{\infty}, \{Z_t\}_{t=0}^{\infty}) = 0,$$

where $\{X_t\}$ collects endogenous variables and $\{Z_t\}$ collects exogenous shocks or policy paths.

Sequence Space vs. State Space

Comparison

Intuition

- State space asks: *what is the current state, and how does it transition?*
- Sequence space asks: *given a shock path, what equilibrium outcome path clears all markets?*

Takeaway

When the distribution is the intractable state variable, sequence space reframes a "huge state" problem as a structured Jacobian problem — amenable to fast, scalable computation.

SSJ Methods

SSJ development

SSJ in Continuous Time

To be developed

Determinacy and Large-Scale Solutions in the Sequence Space

To be developed

2nd SSJ

To be developed

fiscal and monetary policy

To be developed

behavioural macro

To be developed

life-cycle model

To be developed

dynamic spatial model

To be developed

production network

To be developed

het households + het firms

To be developed

het banks

To be developed

optimal policy

To be developed

macro labor

To be developed

Global Solution

Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function

To be developed

DL in sequence space

To be developed

RL in sequence space

To be developed

Reference

Reference

To be developed